

Chapter: Statistics And Probability

Class 12 Business Mathematics (Management) Notes: **Correlation Part**

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1 Introduction to Correlation

Correlation shows us how two things are related to each other. It tells us if one thing changes when the other changes.

There are two important points to understand about correlation:

1. **Direction:** Do both things increase or decrease together, or does one increase while the other decreases?
2. **Strength:** How closely are these changes linked? Is the connection strong or weak?

Why is correlation important?

- It helps us see if two variables are connected.
- It helps us predict one variable by knowing the other.
- It helps us understand relationships in business and daily life.

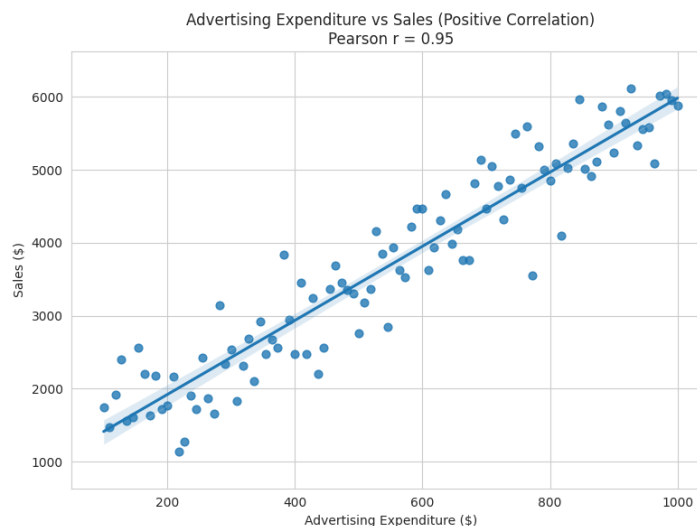
In short, correlation helps us know if two things are connected and how they change together.

1.1 Types of Correlation

1.1.1 Positive Correlation

This occurs when two variables move in the **same direction**. If one variable increases, the other variable also tends to increase. If one decreases, the other also tends to decrease.

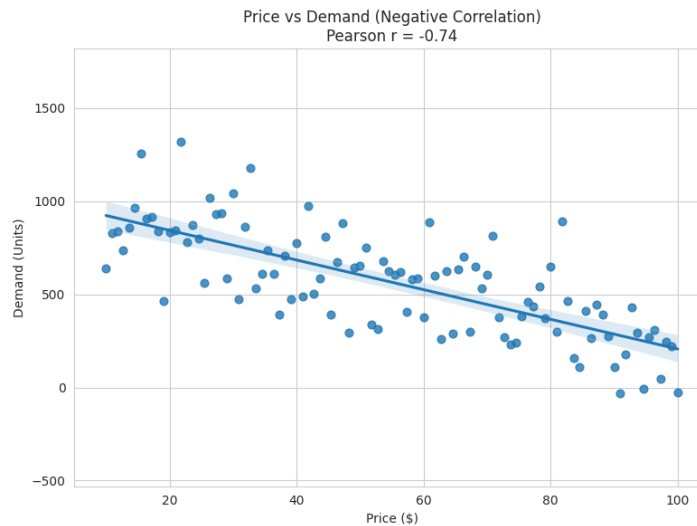
- **Example:** The relationship between **Advertising Expenditure** and **Sales**. Generally, as a company increases its advertising budget, its sales revenue is also likely to increase.



1.1.2 Negative Correlation

This occurs when two variables move in **opposite directions**. If one variable increases, the other variable tends to decrease, and vice versa.

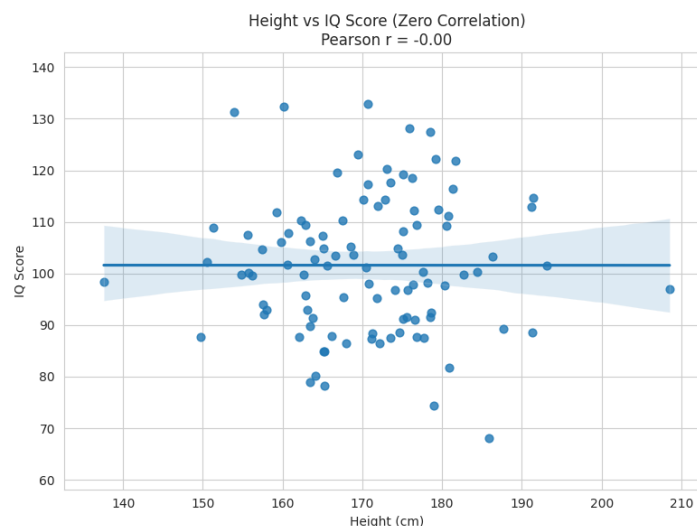
- **Example:** The relationship between the **Price of a Product** and its **Demand**. According to the law of demand, if the price of a commodity increases, the quantity demanded by consumers will typically decrease.



1.1.3 No Correlation (Zero Correlation)

This occurs when there is **no clear linear relationship** between two variables. A change in one variable does not lead to any predictable change in the other.

- **Example:** The relationship between a person's **Height** and their **IQ Score**. There is no logical reason to believe that these two variables are related.



2 Methods for Studying Correlation

To understand the relationship between two variables clearly, we use different methods. These methods help us to:

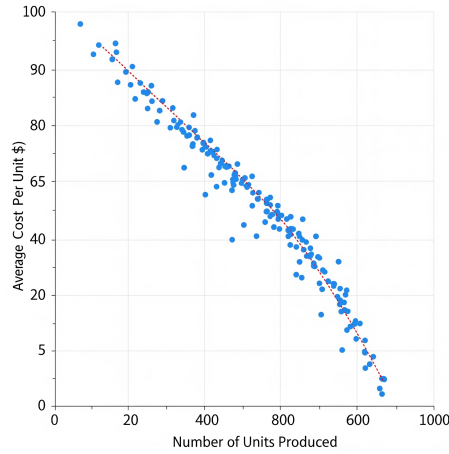
- Visualize how the variables move together,
- Measure the strength and direction of their relationship,
- Make accurate conclusions about their association.

In this chapter, we will learn about two common methods:

1. **Scatter Diagram:** A simple graphical method to see how data points of two variables are related.
2. **Correlation Coefficients:** Mathematical formulas that give us exact numbers showing the strength and direction of correlation.

2.1 Scatter Diagram

A scatter diagram is the simplest method to visualize the relationship. The data points of the two variables (X, Y) are plotted on a graph. By looking at the pattern of the resulting points, one can get a visual idea of the direction and strength of the correlation.



2.2 Karl Pearson's Coefficient of Correlation (r)

This is a mathematical formula that gives a precise numerical value for the strength and direction of a linear correlation. The value, denoted by **r**, always lies between **-1** and **+1**.

- $r = +1 \implies$ Perfect positive linear correlation.
- $r = -1 \implies$ Perfect negative linear correlation.
- $r = 0 \implies$ No linear correlation.
- The closer r is to $+1$ or -1 , the stronger the linear relationship.

Formula using covariance and standard deviation:

Expressed in terms of covariance of X and Y i.e $\text{Cov}(X, Y)$ and its variance i.e $\text{var}(X)$ and $\text{var}(Y)$:

$$r = \frac{\text{Cov}(X, Y)}{\sqrt{\text{var}(X)}\sqrt{\text{var}(Y)}} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

where

$$\text{Cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y})$$

$$\sigma_X = \sqrt{\frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})^2}, \quad \sigma_Y = \sqrt{\frac{1}{N} \sum_{i=1}^N (Y_i - \bar{Y})^2}$$

and $\bar{X}$ and $\bar{Y}$ are the means of X and Y , respectively.

Formula (Product-Moment Method from Raw Data):

$$r = \frac{N \sum XY - (\sum X)(\sum Y)}{\sqrt{N \sum X^2 - (\sum X)^2} \times \sqrt{N \sum Y^2 - (\sum Y)^2}}$$

Worked Example: Calculate Karl Pearson's coefficient for the following data on Advertising Expenditure (X) and Sales (Y).

X (Adv)	Y (Sales)	X^2	Y^2	XY
2	5	4	25	10
3	7	9	49	21
4	8	16	64	32
5	11	25	121	55
6	14	36	196	84
$\sum X = 20$	$\sum Y = 45$	$\sum X^2 = 90$	$\sum Y^2 = 455$	$\sum XY = 202$

Solution:

Given data:

$$N = 5, \quad \sum X = 20, \quad \sum Y = 45, \quad \sum X^2 = 90, \quad \sum Y^2 = 455, \quad \sum XY = 202$$

The formula for Karl Pearson's coefficient is:

$$r = \frac{N \sum XY - (\sum X)(\sum Y)}{\sqrt{N \sum X^2 - (\sum X)^2} \times \sqrt{N \sum Y^2 - (\sum Y)^2}}$$

Substitute the values:

$$r = \frac{5 \times 202 - 20 \times 45}{\sqrt{5 \times 90 - 20^2} \times \sqrt{5 \times 455 - 45^2}} = \frac{1010 - 900}{\sqrt{450 - 400} \times \sqrt{2275 - 2025}} = \frac{110}{\sqrt{50} \times \sqrt{250}}$$

Calculate the square roots:

$$r = \frac{110}{7.071 \times 15.811} = \frac{110}{111.803} \approx 0.9838$$

Conclusion: Since $r = 0.98$ is very close to +1, there is a very strong positive correlation between advertising and sales.

2.3 Spearman's Rank Correlation Coefficient (ρ)

This method is used for **qualitative data** (e.g., beauty, honesty, intelligence) or when the data is not linearly related. It works by assigning ranks to the data values and then measuring the correlation between the ranks.

Formula:

$$\rho = 1 - \frac{6 \sum D^2}{N(N^2 - 1)}$$

where D is the difference between the ranks of the two series ($R_1 - R_2$) and N is the number of pairs.

2.3.1 Case 1: No Tied Ranks

Worked Example: Two judges ranked 5 competitors. Find the rank correlation.

Competitor	Rank by Judge 1 (R_1)	Rank by Judge 2 (R_2)	D = ($R_1 - R_2$)	D^2
A	2	1	1	1
B	1	3	-2	4
C	4	2	2	4
D	5	4	1	1
E	3	5	-2	4
$\sum D^2 =$				14

Solution:

Here, $N = 5$ and $\sum D^2 = 14$.

$$\rho = 1 - \frac{6 \times 14}{5(5^2 - 1)} = 1 - \frac{84}{5(24)} = 1 - \frac{84}{120} = 1 - 0.7 = \mathbf{0.3}$$

Conclusion: A rank correlation of 0.3 indicates a weak positive agreement between the judges.

2.3.2 Case 2: Tied Ranks

If two or more items have the same value, they are given the **average of the ranks** they would have occupied. For each tie, a **Correction Factor (C.F.)** is calculated and added to $\sum D^2$.

The Correction Factor for a tie of size m is calculated as:

$$C.F. = \frac{m(m^2 - 1)}{12}$$

The modified formula for Spearman's rank correlation coefficient, accounting for tied ranks, is:

$$\rho = 1 - \frac{6(\sum D^2 + \sum C.F.)}{N(N^2 - 1)}$$

where:

- N is the number of paired observations,
- $\sum D^2$ is the sum of squared differences of ranks,
- $\sum C.F.$ is the sum of all correction factors for tied ranks.

Worked Example: Two judges ranked 6 competitors with ties in both rankings.

Competitor	Judge 1 Score (X)	Rank (R_1)	Judge 2 Score (Y)	Rank (R_2)
A	90	1	85	1
B	80	2.5	78	4.5
C	80	2.5	78	4.5
D	75	4	78	4.5
E	70	5	70	6
F	65	6	70	6

Explanation:

- In Judge 1's ranking, competitors B and C are tied for 2nd and 3rd places. So, both get average rank:

$$\frac{2 + 3}{2} = 2.5$$

- In Judge 2's ranking, competitors B, C, and D are tied for 4th, 5th, and 6th places. So, each gets average rank:

$$\frac{4 + 5 + 6}{3} = 5$$

Calculate Correction Factors for ties:

- For Judge 1 (two tied items, $m = 2$):

$$C.F._X = \frac{2(2^2 - 1)}{12} = \frac{2 \times 3}{12} = 0.5$$

- For Judge 2 (three tied items, $m = 3$):

$$C.F._Y = \frac{3(3^2 - 1)}{12} = \frac{3 \times 8}{12} = 2$$

Now calculate the differences $D = R_1 - R_2$ and squared differences D^2 :

Competitor	R_1	R_2	$D = R_1 - R_2$	D^2
A	1	1	0	0
B	2.5	5	-2.5	6.25
C	2.5	5	-2.5	6.25
D	4	5	-1	1
E	5	6	-1	1
F	6	6	0	0
$\sum D^2 =$				14.5

Use the modified Spearman's rank correlation formula:

$$\rho = 1 - \frac{6(\sum D^2 + \sum C.F.)}{N(N^2 - 1)} = 1 - \frac{6(14.5 + 0.5 + 2)}{6(36 - 1)} = 1 - \frac{6 \times 17}{210} = 1 - \frac{102}{210} = 1 - 0.4857 = 0.5143$$

Conclusion: The rank correlation coefficient $\rho \approx 0.51$ indicates a moderate positive correlation between the two judges' rankings, accounting for tied ranks properly.